

Cultural Matching and the Persistence of Social Ties

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This paper develops and tests a cultural matching model of social tie persistence using longitudinal ego-network data from the NetSense study. We find that shared cultural tastes serve as a critical mechanism for stabilizing social ties, particularly for weaker relationships, while network opacity—a lack of knowledge regarding an alter’s preferences—independently accelerates tie dissolution.

1 Introduction

A central puzzle in the sociology of culture and social networks is the co-evolution of social ties and cultural tastes. Traditional sociological models of taste formation posit that an individual’s cultural profile is largely a product of their current social network configuration. Tastes are assumed to diffuse through existing homophilous ties, which are themselves structured by socio-demographic similarity—“status homophily” (Lazarsfeld and Merton, 1954; McPherson et al., 2001). Under this view, “birds of a feather flock together,” and due to this mutual flocking, they end up consuming the same cultural goods (Mark, 1998b, 2003). Tastes are construed as highly malleable, “thin” contents liable to shift whenever the surrounding network ties change. Yet, longitudinal network research reveals that personal networks are constantly in flux, with ties constantly being added, modified, and deleted (Burt, 2000, 2002; Wellman et al., 1997). This empirical fact of network volatility raises a major question: if networks are unstable and tastes are merely thin, network-contingent contents, how do we account for the durable, resilient cultural profiles that individuals maintain over time (Bourdieu, 1984; Holt, 1998)?

In this paper, we develop and test a **cultural matching model** of tie persistence. We propose that instead of tastes merely adapting to match pre-existing networks, pre-existing cultural dispositions serve as an active “social glue” that determines which network ties survive and which dissolve. Under this model, dyadic cultural matching (commonalities in tastes and activities) increases the over-time stability and cognitive salience of social ties, net of usual dyadic determinants such as subjective closeness and tie duration. Using rich longitudinal ego-network data from the NetSense study, we track the persistence of social ties across multiple

college-year transitions. To isolate peer dynamics, we restrict our focus to non-kin, on-campus student ties. We unpack this cultural matching mechanism in two ways. First, we establish the baseline protective effect of aggregate cultural matching on tie persistence. Second, we explore the role of “network opacity”—the degree to which an ego lacks knowledge of their alter’s preferences (indicated by “Don’t Know” responses)—as an independent predictor of tie decay.

2 Theoretical Framework

2.1 Tastes and Social Networks: Two Perspectives

The relationship between cultural tastes and social networks has been conceptualized in two contrasting ways in sociological theory. The **network transmission perspective**, drawing on constructural imagery (Carley, 1991), views tastes as malleable, network-contingent contents (Erickson, 1988, 1996; Mark, 1998a, 2003). Recent work suggests that this malleability is constrained by deeper, latent cultural structures. In this view, social ties constitute the pre-existing “infrastructure” or “pipes” through which cultural practices diffuse (Borgatti, 2005; Erickson, 1996). Tastes are learned, adopted, and abandoned based on continuous exchange of cultural information and local bandwagon imitation within an actor’s immediate ego network (Mark, 2003). Because personal networks exhibit substantial over-time volatility—with 30% to 60% turnover annually in most personal networks (Morgan et al., 1997; Suitor et al., 1997)—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the **cultural capital perspective** conceptualizes tastes as highly durable, embodied dispositions reaped from early socialization (Bourdieu, 1984, 1990; Holt, 1997, 1998). Emphasizing the “interconvertibility” of different forms of capital (Bourdieu, 1986), this view sees cultural aptitudes not merely as by-products of ties, but as fundamental resources that help people form and sustain network connections. Here, networks are seen as fleeting structures built and maintained via interactions (DiMaggio, 1987), and individuals actively use pre-existing tastes to construct novel relationships or shed culturally incompatible ties (Vaisey and Lizardo, 2010). If tastes are durable and personal networks are volatile, it is highly probable that tastes serve as *drivers* of network connections rather than being exclusively determined by them. This leads to the concept of **choice homophily** or cultural matching: new social connections are selected, and existing connections are maintained, based on pre-existing cultural compatibility (Lazarsfeld and Merton, 1954; Werner and Parmelee, 1979). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact selection keyed around cultural alignment, rather than homophily acting as an exogenous structural constraint (Carley, 1991; DiMaggio, 1987).

2.2 Cultural Matching and Tie Persistence

A major challenge for any newly formed social relationship is what Burt (Burt, 2000), drawing on Stinchcombe (Stinchcombe, 1965), refers to as the “**liability of newness.**” Newly formed dyadic ties are fragile and exhibit a high baseline hazard of decay (“tie decay”), which steeply

declines as a relationship matures and ripens into close friendship. What stabilizes a tie and helps it overcome this liability of newness? While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we propose that **cultural matching** serves as a vital stabilizing mechanism. Shared tastes and cultural practices provide a common cognitive and behavioral baseline that facilitates smooth, low-cost social interaction (Collins, 2004; DiMaggio, 1987).

When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the ego’s active network (Burt, 2000). Therefore, we expect that commonalities in cultural tastes and leisure practices will significantly reduce the hazard of tie decay, net of subjective closeness and structural controls (**The Baseline Matching Hypothesis**).

2.3 Network Opacity

Finally, we introduce the concept of **network opacity**. If shared cultural knowledge stabilizes ties, then a lack of mutual cultural knowledge—network opacity—is a diagnostic signal of relationship precarity. When an ego reports “Don’t Know” regarding an alter’s preferences across multiple cultural domains, this indicates a lack of conversational intimacy and shared experiences, which should independently predict a higher hazard of tie dissolution (Burt, 2002; Wellman et al., 1997). We test both the main effect of network opacity on tie decay and whether opacity moderates the protective effect of known cultural matches.

3 Data and Methods

Data for this analysis comes from the NetSense study, a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students’ social networks, demographic backgrounds, and cultural preferences. A comprehensive codebook, data dictionary, and repository for the NetSense project are available at <https://olizardo.github.io/NetSense-Codebook/>.

We collected information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended activity nominations. For the closed-form items, egos were asked to rate both their own and their alters’ level of interest across six broad cultural and leisure domains: music, movies, books, sports, games, and outdoor activities. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity.

3.1 Study Waves

The NetSense study is a longitudinal survey of college students collected over multiple waves. For this analysis of tie persistence, we pool data across adjacent waves to construct a person-period dataset. To isolate peer dynamics, we restrict our sample to non-kin, on-campus ties (i.e., alters who are also students at Notre Dame). To capture relationship dynamics across the

entirety of the college experience, we construct a discrete-time person-period dataset spanning all available adjacent survey waves (Wave 1 through Wave 8). This design allows us to observe tie persistence continuously across varying intervals of the students’ academic careers, capturing both the initial network churn of freshman year and the consolidation of networks approaching graduation.

3.2 Measures

3.2.1 Dependent Variable

Tie Persistence: A binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2), and 0 if the tie has dissolved (i.e., the alter is no longer listed in the ego’s network). To determine tie presence, egos were asked to list the people with whom they interact most frequently, with the prompt: “Please list the people with whom you have spent the most time in the last month.” Egos could nominate up to 20 alters in each wave.

3.2.2 Independent Variables

- *Closed-Form Cultural Matching*: A count variable (ranging from 0 to 6) indicating the number of closed-form cultural domains in which the ego and alter share similar tastes. Egos were asked to rate their own interest and their alters’ interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey prompt for egos was: “How interested are you in the following activities?” and for alters: “How interested is this person in the following activities?” Response categories included “Very interested,” “Somewhat interested,” and “Not at all interested.” A domain was scored as a “match” if the ego rated both themselves and their alter identically.
- *Open-Ended Activity Matching*: A count variable (ranging from 0 to 5) capturing the number of shared specific activity nominations. Egos were asked to freely list up to five leisure activities they enjoyed with the prompt: “Please list up to 5 specific activities that you enjoy doing in your free time.” Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed, prompted with: “Which of your favorite activities does this person also enjoy?”
- *Cultural Network Opacity*: A count variable (ranging from 0 to 6) measuring the number of cultural domains for which the ego responded “Don’t know” regarding their alter’s preferences. For the closed-form domain questions regarding alters (“How interested is this person in the following activities?”), respondents were provided a fourth response category: “Don’t know.” This variable sums the number of domains where this option was selected.

3.2.3 Control Variables

- *Subjective Closeness*: Measured with the prompt, “How close are you to this person?” Response categories were “Not close” (1), “Somewhat close” (2), and “Very close” (3). This is treated as a factor variable in the models.
- *Relationship Type*: Derived from the prompt “How are you connected to this person?” We include binary indicators for whether the tie is a **Friend** (vs. other types) or a **Roommate/Dormmate** (capturing on-campus structural proximity).
- *Frequency of Interaction*: Measured with the prompt, “How often do you communicate with this person?” Responses were dichotomized into a **High Frequency** indicator (**Daily** or **Weekly**) versus the reference category (**Monthly** or **Less often**).
- *Tie Duration*: A continuous measure of how long the ego has known the alter (in years), asked via “How long have you known this person?” We include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.
- *Demographics*: We control for ego gender and alter gender (1 = Woman, 0 = Man), along with a multiplicative interaction term to capture same-gender vs. cross-gender dynamics. We also include a measure of *Race Homophily*, constructed by matching self-reported race for the ego and self-reported race for the alter. Ties are classified as **Homophilous** (same race), **Heterophilous** (different race), or **Unknown** (if either ego or alter race was missing).
- *Time Period*: Dummy variables for each wave transition interval (e.g., Wave 1 to 2, Wave 2 to 3, etc.) to allow the baseline hazard of tie persistence to vary across the panel.

3.3 Analytical Strategy and Model Specification

Because our data tracks relationships over time and individuals can nominate multiple alters, our data structure consists of dyad-periods nested within egos. To properly analyze tie persistence and account for right-censoring, we employ a discrete-time survival analysis framework (event history modeling). Since the outcome variable—whether a tie persists into the next wave—is binary, we estimate the hazard of tie survival using mixed-effects logistic regression.

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), we include random intercepts for egos. This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1)) = \beta_0 + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where Y_{ijt} is the indicator of tie persistence for ego i and alter j at time t , X_{ijt} represents the vector of our core theoretical predictors (cultural matching and network opacity), C_{ijt} is a vector of dyadic and structural control variables (including time period indicators to allow the

baseline hazard to vary), and u_i is the ego-specific random intercept, assumed to be normally distributed with mean zero.

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package in R, utilizing the BOBYQA optimizer to ensure model convergence. In models interacting subjective closeness with our cultural variables, we calculate and plot average marginal effects to facilitate interpretation of the non-linear interaction terms.

For a full summary of descriptive statistics, see Tables ?? and ?? in the Appendix.

4 Results

Our analysis proceeds in two stages. First, we examine the main effects of cultural matching and network opacity on the likelihood of tie persistence, holding dyadic and structural factors constant. Second, we explore how the role of cultural matching and network opacity varies across different levels of baseline tie strength (subjective closeness).

4.1 Main Effects: Cultural Matching and Network Opacity

Table ?? presents the odds ratios from a series of mixed-effects logistic regression models predicting tie survival. Model 1 establishes our baseline, demonstrating that **closed-form cultural matching** significantly increases the odds of tie persistence. For each additional broad cultural domain (e.g., music, movies, sports) in which an ego and alter share tastes, the tie is substantially more likely to survive into the subsequent wave, net of controls for subjective closeness, tie duration, frequency of interaction, and demographic homophily.

In Model 2, we introduce **open-ended activity matches**. The inclusion of this variable demonstrates that specific, shared behavioral practices (such as co-participating in a particular hobby or sport) provide additive protective value above and beyond broad categorical taste similarities. Both closed-form and open-ended matching independently reduce the hazard of tie decay.

Model 3 tests the independent effect of **network opacity**—measured as the number of domains in which the ego reported “Don’t Know” regarding their alter’s preferences. As expected, network opacity is a strong negative predictor of tie persistence. A lack of mutual cultural knowledge signals a lack of conversational intimacy; as the number of “unknown” domains increases, the likelihood of tie survival sharply drops.

To contextualize these coefficients, Figure 1 displays the average marginal effects of moving from the minimum to the maximum observed values for each of our core theoretical variables. Across the board, cultural matching serves as a robust protective mechanism against tie decay, whereas network opacity acts as a profound liability.

4.2 The Moderating Role of Subjective Closeness

While the main effects demonstrate that cultural alignment generally stabilizes ties, we next assess whether this “social glue” is uniformly necessary for all relationships. We estimated auxiliary interaction models (not shown in tables, but visualized via marginal effects) to test

whether the protective effects of cultural matching and the detrimental effects of network opacity are moderated by baseline subjective closeness.

Because interaction terms in logistic models can be opaque due to the non-linear link function, Figure 2 plots the predicted average marginal effect of our three cultural variables across ties rated as “Not Close,” “Somewhat Close,” and “Close.”

The results reveal a striking asymmetry: the role of cultural matching and opacity is highly dependent on tie strength. For weak ties (those rated “Not Close”), the marginal effects are steepest. For these vulnerable relationships, an increase in shared cultural tastes—whether measured via closed-form domains or open-ended activities—dramatically increases the probability of tie persistence. Conversely, for strong ties (those rated “Close”), the marginal effect slopes are essentially flat.

This interaction suggests that shared cultural tastes provide the critical *scaffolding* necessary to help weak, budding ties survive their initial liability of newness. Strong ties, however, are deeply institutionalized and anchored by other relational mechanisms (such as deep emotional attachment or multiplex obligations), making them resilient to decay regardless of fluctuations in cultural matching. Similarly, network opacity exerts a severe negative toll on weak ties, but close friendships are robust enough to withstand a degree of ignorance regarding specific cultural domains.

5 Discussion and Conclusion

5.1 Summary of Key Findings

In this paper, we sought to address a central puzzle in the sociology of culture and social networks: how do cultural tastes shape the evolutionary dynamics of personal networks? Using longitudinal ego-network data from the NetSense study, we tested a cultural matching model of tie persistence. Our results provide robust evidence that shared cultural tastes are a vital mechanism for stabilizing social ties. Specifically, both broad, closed-form cultural matches and specific, open-ended activity matches independently reduce the hazard of tie dissolution. Furthermore, we identified network opacity—the lack of mutual cultural knowledge—as a powerful, independent liability that accelerates tie decay. Crucially, the protective effect of cultural matching and the detrimental effect of network opacity are heavily concentrated among weak ties. For subjectively close ties, the underlying cultural scaffolding becomes less relevant to tie survival.

5.2 Theoretical and Substantive Interpretations

These findings carry significant theoretical implications for how we understand the relationship between tastes and networks. The results challenge the assumption of the pure network transmission perspective, which treats cultural tastes merely as fluid by-products of existing social ties (Mark, 2003). Instead, our findings align more closely with the cultural capital perspective (Bourdieu, 1984; Vaisey and Lizardo, 2010), suggesting that tastes function as

durable, portable resources that individuals actively use to select, maintain, and discard social relationships.

We conceive of cultural matching as an essential form of “social scaffolding.” New and weak ties face a high liability of newness (Burt, 2000). In the absence of deep emotional bonds, multiplex obligations, or institutional constraints, shared cultural tastes provide a low-friction baseline for interaction. Shared music, movies, or sports generate immediate conversational fodder and lower the cognitive costs of coordination. Over time, as a tie matures and institutionalizes into a close friendship, this scaffolding is no longer strictly necessary; the relationship becomes resilient on its own terms. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles: people are constantly churning through ties, but they use their stable cultural compass to determine which ties stick.

5.3 Limitations and Future Research

Despite these robust findings, several limitations point toward fruitful avenues for future research. First, our empirical focus on a sample of college students at a single university limits the generalizability of our claims. The college transition is a period of heightened network flux and identity formation. Future research should examine whether cultural matching operates similarly as a stabilizing force in adult populations, workplace settings, or among older adults where network dynamics and the salience of different cultural domains may differ.

Second, our current operationalization treats all cultural matches as theoretically equivalent. Future research could disaggregate these domains to test whether high-status cultural matches (e.g., classical music, literature) confer different relational advantages compared to popular or low-status domains (DiMaggio, 1987). Additionally, exploring the valence of cultural matching—such as shared *dislikes* (Erickson, 1996)—could provide a more nuanced picture of relational boundary-drawing.

Finally, while our discrete-time survival framework effectively models tie decay, future studies would benefit from deploying stochastic actor-oriented models (SAOMs) to simultaneously model tie creation, tie dissolution, and individual taste change. Such an approach could fully disentangle the reciprocal processes of network-based cultural influence and culture-based social selection.

Figures

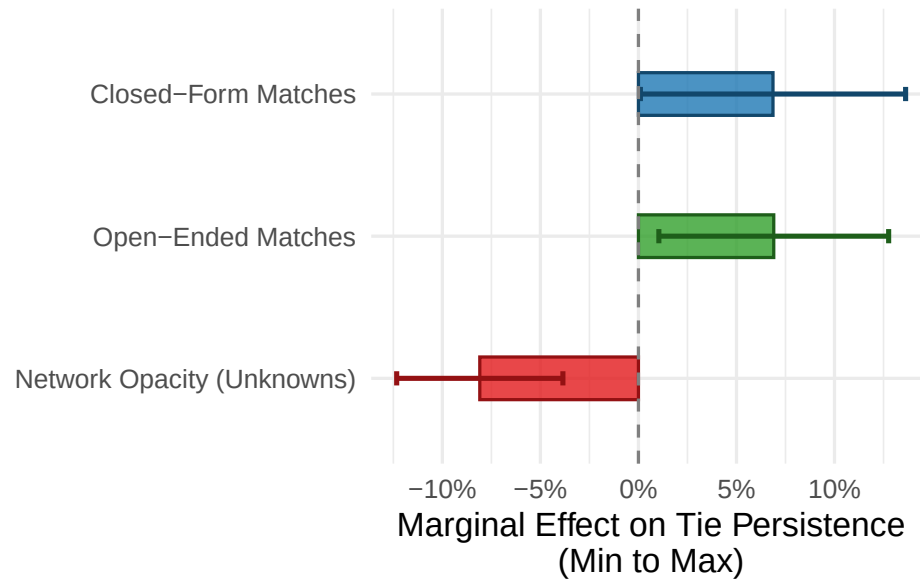


Figure 1: Marginal Effects of Cultural Matching and Network Opacity on Tie Persistence. Note: Estimates represent the average marginal effect on the probability of tie persistence.

Tables

(Tables inserted dynamically via Quarto/R)

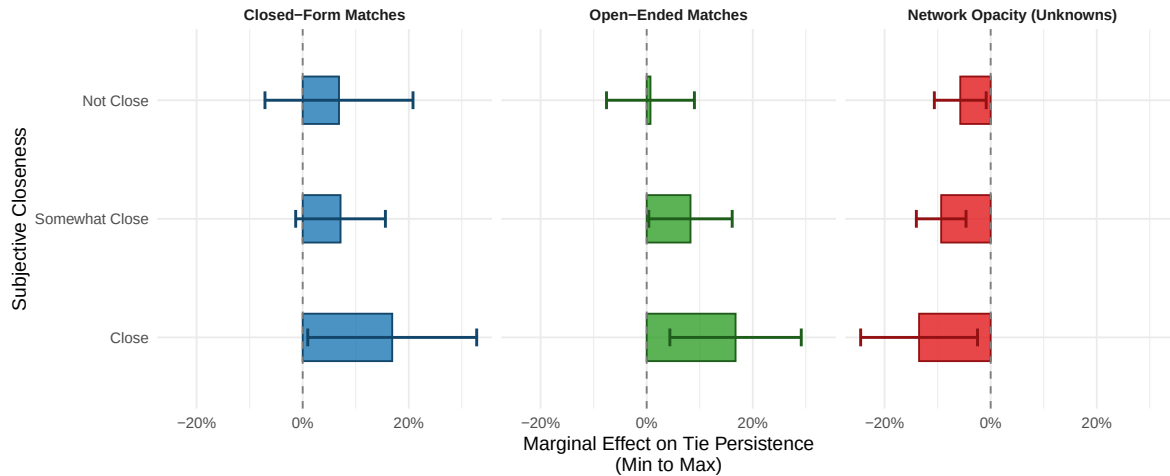


Figure 2: Marginal Effects of Cultural Matching and Network Opacity on Tie Persistence by Subjective Closeness. Note: Estimates represent the average marginal effect on the probability of tie persistence.

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